

Face Recognition App

1-Live Camera recognition face_rec.py

1-INPUT : **database_students.csv**
contains name,age,records etc ...
That are used to recognise faces in
live Camera app

2-PROCESSING :
keras_data_augmentation.py :
from 1 image we obtain 10
different images (zoom, rotations
etc...) --> This data is useful to
implement CNN score and
accuracy assessment in face
recognition backend script

3-OUTPUT : **Live Image Captures
from Webcam** :
NAME_SURNAME_DATE%**HOURL**.jpg

a-Building Entrance & Exit log :
Statistics database
database_logs.csv hourly, daily,
monthly etc ...

2-CNN Face Recognition face_rec_keras.py

1-INPUT : Augmented images from step 1.2
using the script **keras_data_augmentation.py**

2-OUTPUT : Face Alignment,
Embedding vectors, distance
threshold measurement,
histograms, KNN and SVM
accuracy calculations, t-SNE data
visualisation.

3-PowerBI Data Visualisation

3-INPUT : Database created in step
1.3.a and parsed into
database_logs_parsed.csv
containing the entrance and exit
logs : hourly, daily or monthly logs
for each student.